Loan Monitoring System: Development Plan and Acceptance Tests

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Stack: Python (FastAPI) · MongoDB · React (admin web) · React Native (member mobile app)
Version: 1.0

1. Purpose and Scope

This document defines how the Loan Monitoring System will be built and how it will be accepted. It covers:

- **Member Mobile App** (Android and iOS): registration, loan applications, balances, payment history, schedules, reminders and notifications.
- **Admin Web System** (computer browser): staff access control, member management, loan products, loan processing, payment posting, overdue monitoring, reports and transaction history.
- **System Management**: secure database, real-time synchronization, backup and recovery, audit trail and a user-friendly interface.

Out of scope for version 1, available as change requests: online payments (GCash/Maya/bank gateway), SMS gateway integration, e-signature on loan agreements, accounting/general ledger, multi-branch consolidation and credit bureau (CIC) submission.

> **Note on sample figures.** Interest rates, penalties, fees and reminder schedules in this document are sample settings that make the tests concrete. They will be replaced with the company's actual loan policies in Phase 1. Every one of them is configurable in the admin system.

2. Architecture

Layer	Technology	Notes
Member Mobile App	React Native (Expo), TypeScript	One codebase for Android 8+ and iOS 15+. Secure token storage; optional fingerprint/Face ID unlock
Admin Web System	React 18 + Vite, TypeScript, React Router, TanStack Query, Ant Design	Runs in Chrome, Edge and Firefox on office computers; no installation
API Server	Python 3.12, FastAPI, Pydantic v2, Uvicorn/Gunicorn	REST JSON API with OpenAPI docs; single API for both apps
Database	MongoDB 7 as a replica set (MongoDB Atlas or self-hosted)	A replica set is required for multi-document transactions on loan/payment postings
Data access	Motor (async driver) + Beanie ODM	Schema validation in Pydantic plus MongoDB JSON-schema validators on key collections
Background jobs	APScheduler, or Celery + Redis	Daily overdue/penalty run, reminder dispatch, report generation, backups
Real-time sync	WebSocket channel (admin) + push notifications (mobile) + cache refresh on app open	A payment posted by staff updates the member app within seconds

Layer	Technology	Notes
Push notifications	Expo Push Service (FCM for Android, APNs for iOS)	
File storage	S3-compatible object storage (AWS S3 or DigitalOcean Spaces)	Private bucket, short-lived signed URLs for IDs and documents
Reports	pandas + openpyxl (Excel), WeasyPrint (PDF)	
Hosting	Linux VPS, Docker Compose, Nginx reverse proxy, Let's Encrypt SSL	Staging and production environments
CI/CD	GitHub Actions	Lint, type check, tests, build, deploy to staging
Monitoring	Sentry (errors), Uptime monitor, structured JSON logs	

2.1 Key design decisions

1. **Money is stored as integer centavos (int64)** in MongoDB, for example ₱4,727.98 is stored as `472798`. All calculations use Python `Decimal` with ROUND_HALF_UP to the centavo. Floats are never used for money.
2. **Financial postings use MongoDB transactions.** Loan release, payment posting and payment voiding update the loan, its installments, the payment record and the audit log in one atomic transaction. If any step fails, nothing is saved.
3. **Optimistic locking** on loans (`version` field) prevents two cashiers from posting against the same loan at the same moment and corrupting balances.
4. **Loan product terms are snapshotted** into each loan at release. Changing a product later never changes existing loans.
5. **Append-only ledger.** Payments are never edited or deleted. Corrections are made by voiding (with a reason) and re-posting.
6. **Append-only audit log** for every create, update, approve, release, post, void and login event.
7. **Timezone:** all timestamps are stored in UTC and displayed in Asia/Manila. Due dates are calendar dates.

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3. Users, Roles and Permissions

Permission	Admin	Manager	Loan Officer	Cashier	Auditor	Member (app)
Manage staff users and roles	✔					
Configure loan products, penalties, settings	✔	✔				
Register and edit members	✔	✔	✔			Own profile (key fields need approval)
Approve member registration from app	✔	✔	✔			
Encode and review loan applications	✔	✔	✔			Apply and track own
Approve or decline loans	✔	✔				
Release loans	✔	✔				
Post payments	✔	✔		✔		
Void payments	✔	✔				

Permission	Admin	Manager	Loan Officer	Cashier	Auditor	Member (app)
View monitoring dashboard	✔	✔	✔	✔	✔	
Generate and export reports	✔	✔	✔	✔ (collections only)	✔	
View audit trail	✔	✔			✔	
View own loans, payments and schedule						✔

Segregation of duties (configurable, on by default): the staff member who encoded a loan application cannot approve it.

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4. Data Model (MongoDB Collections)

Collection	Key fields	Indexes
`staff_users`	name, email, password_hash (Argon2), role, status, failed_logins, locked_until, last_login_at	unique email
`members`	member_no, first/middle/last name, birthdate, sex, civil_status, address, mobile, email, employment/income source, gov_id {type, number, file_key}, photo_key, status (pending, active, suspended, closed), app_account {username, password_hash, verified_at}, consent {version, accepted_at}, created_by	unique member_no; unique mobile; text index on names
`loan_products`	code, name, interest_rate, rate_period (monthly/annual), method (diminishing, add_on), terms_allowed, frequency (weekly, semi_monthly, monthly), min/max amount, fees [{name, type: percent/fixed, value, deducted_from_proceeds}], penalty {rate, basis, grace_days}, max_active_loans, active	unique code
`loan_applications`	application_no, member_id, product_id, amount, term, frequency, purpose, documents[], channel (mobile, walk_in), status (submitted, under_review, approved, declined, cancelled, released), status_history [{status, by, at, remarks}], encoded_by, approved_by	unique application_no; (member_id, status)
`loans`	loan_no, application_id, member_id, product_snapshot, principal, total_interest, fees[], net_proceeds, release_date, first_due_date, maturity_date, status (active, past_due, fully_paid, restructured, written_off), balances {principal, interest, penalty, total}, days_past_due, version	unique loan_no; (member_id, status); status
`installments`	loan_id, seq, due_date, principal_due, interest_due, penalty_due, principal_paid, interest_paid, penalty_paid, status (unpaid, partial, paid, overdue), paid_at, last_penalty_assessed_at	(loan_id, seq) unique; (due_date, status)
`payments`	receipt_no (OR number), loan_id, member_id, amount, method (cash, bank, e_wallet, check), reference_no, payment_date, allocation [{installment_seq, penalty, interest, principal}], advance_amount, received_by, status (posted, voided), void {reason, by, at}	unique receipt_no; (loan_id, payment_date); payment_date
`notifications`	member_id, type, title, body, data, channels, sent_at, read_at	(member_id, created_at)
`devices`	member_id, push_token, platform, last_seen_at	unique push_token
`audit_logs`	actor {type, id, name}, action, entity, entity_id, before, after, ip, user_agent, at	(entity, entity_id); at; actor.id
`settings`	company profile, reminder schedule, quiet hours, payment allocation order, segregation_of_duties, password policy	singleton

Collection	Key fields	Indexes
`counters`	name, seq	Used for member_no, loan_no, application_no

Retention: members and loans are never hard-deleted; they are closed or archived. Audit logs are read-only to every role.

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5. Business Rules

5.1 Amortization

- **Diminishing balance:** fixed installment `A = P × r ÷ (1 − (1 + r)^−n)`, where `r` is the rate per period. Interest per period = outstanding principal × r. The last installment absorbs any rounding difference so the principal ends at exactly ₱0.00.
- **Add-on (flat):** total interest = P × r × n. Installment = (P + total interest) ÷ n. Principal and interest are spread evenly, and the last installment absorbs rounding.
- **Due dates:** the first due date is one period after the release date, unless set manually at release. If a day-of-month does not exist (for example the 31st), the last day of that month is used.

5.2 Fees and proceeds

- Fees are computed at release. Fees marked "deducted from proceeds" reduce the net cash released. Other fees are added to the first installment.

5.3 Payment allocation

- Default order per installment: **penalty → interest → principal**, oldest unpaid installment first.
- Amounts beyond all currently due installments are applied to the next installments in order, as an advance payment. A setting can apply excess to principal instead.
- A payment greater than the total outstanding balance is rejected.
- The loan becomes **fully paid** when all balances reach ₱0.00. The member is notified, and a fully paid notice can be printed.

5.4 Overdue and penalty

- An installment becomes **overdue** the day after its due date if not fully paid.
- A **penalty** is assessed after the grace period: penalty rate × unpaid installment amount (principal + interest) for each 30-day period, or part of a period, past due. Sample setting: 3% per month, 3 grace days.
- A daily job runs at 00:30 Asia/Manila to update statuses, days past due, penalties and aging buckets. The job is **idempotent**: running it twice on the same day does not add a second penalty.
- **Aging buckets:** Current, 1–30, 31–60, 61–90 and 90+ days past due.
- Staff with Manager role can waive a penalty with a required reason, which is audited.

5.5 Loan application and eligibility

- Only **active** members can apply.
- The amount must be within the product's minimum and maximum, and the term must be one the product allows.
- A member cannot exceed the product's `max_active_loans`.
- Status flow: submitted → under review → approved / declined → released. A member can cancel only while the status is submitted.

5.6 Reminders

- Sample schedule: 3 days before the due date, on the due date, 1 day after, then every 7 days while overdue.
- Reminders are sent only between 8:00 AM and 8:00 PM Asia/Manila. A reminder is never sent twice for the same installment and trigger.

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6. API Outline (FastAPI, `/api/v1`)

Area	Endpoints
Staff auth	`POST /auth/staff/login`, `POST /auth/refresh`, `POST /auth/logout`, `POST /auth/staff/change-password`
Member auth	`POST /auth/member/register`, `POST /auth/member/verify-otp`, `POST /auth/member/login`, `POST /auth/member/forgot-password`
Staff users	`GET/POST /staff-users`, `PATCH /staff-users/{id}`, `POST /staff-users/{id}/unlock`
Members	`GET/POST /members`, `GET/PATCH /members/{id}`, `POST /members/{id}/approve`, `POST /members/{id}/suspend`, `GET /members/{id}/ledger`
Loan products	`GET/POST /loan-products`, `PATCH /loan-products/{id}`
Calculator	`POST /loan-products/{id}/preview-schedule`
Applications	`GET/POST /loan-applications`, `GET /loan-applications/{id}`, `POST /…/{id}/review`, `/approve`, `/decline`, `/cancel`
Loans	`POST /loan-applications/{id}/release`, `GET /loans`, `GET /loans/{id}`, `GET /loans/{id}/schedule`, `POST /loans/{id}/waive-penalty`
Payments	`POST /payments`, `GET /payments`, `GET /payments/{id}`, `POST /payments/{id}/void`, `GET /payments/{id}/receipt.pdf`
Dashboard	`GET /dashboard/summary`, `GET /dashboard/aging`, `GET /dashboard/due-today`
Reports	`GET /reports/{type}?from=&to=&format=pdf
Member app	`GET /me`, `PATCH /me`, `GET /me/loans`, `GET /me/loans/{id}`, `GET /me/payments`, `GET/POST /me/applications`, `GET /me/notifications`, `POST /me/notifications/{id}/read`, `POST /me/devices`
Audit	`GET /audit-logs`
Real-time	`WS /ws/admin` (events: payment.posted, application.submitted, loan.status_changed)
Health	`GET /health`

All write endpoints that post money accept an `Idempotency-Key` header, so a double-click or network retry never posts a payment twice.

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7. Screens

7.1 Member Mobile App

1. Welcome, Login, Register (personal info, ID upload, selfie optional, consent), OTP verification, Forgot password
2. Home: total outstanding balance, next due date and amount, overdue banner, quick actions

3. My Loans: list, then Loan Detail (balance breakdown, progress, amortization schedule, remaining installments)
4. Payment History: list with filters, then Payment Detail (OR number, allocation)
5. Apply for a Loan: choose product, amount, term, preview schedule, upload documents, submit
6. My Applications: status timeline
7. Notifications inbox
8. Profile and settings: contact details, change password, biometric unlock, logout

7.2 Admin Web System

1. Login, Change password
2. Dashboard: portfolio outstanding, released this month, collections today/month, due today, overdue count and amount, aging chart, recent activity (live)
3. Members: list/search, Member profile (info, documents, loans, ledger), Pending app registrations
4. Loan Products: list and editor with schedule preview
5. Applications: queue by status, Application detail (review, approve/decline, release)
6. Loans: list with filters, Loan detail (schedule, payments, penalties, waive)
7. Payments: Post payment (search member/loan, amount, OR number, allocation preview), Payment list, Void
8. Overdue Monitoring: past-due list by aging bucket, with contact details and export
9. Reports
10. Staff Users and Roles
11. Settings
12. Audit Trail

7.3 Reports (PDF and Excel)

Loan releases · Collections (daily/period, by cashier) · Outstanding balances · Past due / aging · Member ledger · Loan amortization schedule · Payment history / transaction history · Fully paid loans · Penalty assessed and waived · Official receipt reprint.

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8. Non-Functional Requirements

Area	Requirement
Security	HTTPS only; Argon2 password hashing; JWT access tokens (15 min) with rotating refresh tokens; account lockout after 5 failed logins for 15 minutes; rate limiting on auth endpoints; RBAC enforced on the server for every endpoint; input validation; private file storage with signed URLs; secrets in environment variables
Data privacy	Aligned with the Data Privacy Act of 2012 (RA 10173): consent captured at registration, least-privilege access, audit trail, encrypted backups, privacy notice in app
Performance	API p95 response under 500 ms for normal operations with 50,000 members and 200 concurrent users; reports for one month of data generated under 30 seconds
Real-time	Payment posted by staff visible in the member app within 10 seconds when the app is open; push notification within 1 minute
Availability	99.5% monthly target; scheduled maintenance announced in advance
Backup and recovery	Automated daily encrypted backup kept for 30 days, plus weekly kept for 12 weeks, stored off-server; RPO 24 hours (or near zero with Atlas continuous backup); RTO 4 hours; restore tested before go-live
Usability	Mobile app usable by first-time smartphone users; large tap targets; English labels (Filipino optional); every list searchable; destructive actions require confirmation
Compatibility	Android 8.0+, iOS 15+; Chrome, Edge, Firefox (latest two versions); admin screens from 1366×768

Area	Requirement
Maintainability	Type hints and linting (ruff, mypy, ESLint, Prettier); automated tests in CI; README and deployment runbook

9. Repository Structure

```
loan-monitoring/
├── backend/                # Python FastAPI
│   ├── app/
│   │   ├── api/v1/         # routers: auth, members, products, applications, loans, payments,
reports, me
│   │   ├── core/           # config, security, rbac, logging
│   │   ├── models/         # Beanie documents
│   │   ├── schemas/        # Pydantic request/response models
│   │   ├── services/       # amortization.py, allocation.py, penalty.py, posting.py,
notifications.py, reports/
│   │   ├── jobs/           # daily_overdue.py, reminders.py, backup.py
│   │   └── main.py
│   ├── tests/              # unit/, integration/, acceptance/
│   └── pyproject.toml
├── admin-web/              # React + Vite + TypeScript
├── member-app/             # React Native (Expo) + TypeScript
├── infra/                  # docker-compose.yml, nginx/, backup scripts, GitHub Actions
└── docs/                   # this plan, API docs, runbook, user guides
```

10. Delivery Plan (10 Weeks)

Phase	Weeks	Key tasks	Exit criteria
1. Requirements and design	1	Confirm loan products, interest methods, penalty and fee rules, OR numbering, roles, report formats; collect sample forms and a real amortization sheet; UI wireframes for app and admin; finalize data model	Client sign-off on business rules sheet and wireframes; sample figures in Section 5 replaced with actual policies
2. Foundation	2	Repos, CI, Docker, staging server, MongoDB replica set; staff auth, RBAC, audit log middleware; admin shell; member app shell	Staff can log in on staging; RBAC tests pass
3. Members and products	3	Member CRUD, documents upload, member app registration + OTP + approval; loan products and schedule preview; amortization engine with unit tests	AT-MEM and AT-PRD tests pass; amortization matches the client's sample sheet to the centavo
4. Applications and release	4	Application flow (admin and app), approval with segregation of duties, release with fees and schedule generation, loan detail	AT-APP and AT-REL tests pass
5. Payments and monitoring	5	Payment posting with allocation, OR numbers, idempotency, voiding, receipts; dashboard; daily overdue/penalty job; aging	AT-PAY and AT-OVD tests pass; **Admin System demo to client (payment milestone 2)**
6. Member app features	6–7	Home, loans, schedule, payment history, applications, notifications inbox, profile, biometric unlock; push notifications and reminders; real-time updates	AT-MOB, AT-NTF and AT-SYNC tests pass on Android and iOS test builds
7. Reports and hardening	8	All reports in PDF and Excel; audit trail screen; backups and restore drill; security review; performance test with seeded 50,000 members	AT-RPT, AT-BKP, AT-SEC and AT-PERF tests pass

Phase	Weeks	Key tasks	Exit criteria
8. UAT	9	Client staff run the acceptance tests on staging with realistic data; fix defects; data migration of existing loans if needed	All Critical and High defects closed; UAT sign-off
9. Go-live and training	10	Production deployment, app store submissions, staff training (admin, cashier, manager), user guides, handover of source code and credentials	Production live; apps published or in store review; **turnover (final milestone)**

10.1 Definition of Done (each feature)

Code reviewed and merged · unit and integration tests written and passing in CI · RBAC applied on the server · audit log entries created for every state change · works on staging · related acceptance tests pass · no open Critical/High defects.

10.2 Defect severity

Severity	Definition	UAT gate
Critical	Wrong money computation, data loss, security breach, system unusable	Must be fixed before go-live
High	Core feature blocked with no workaround	Must be fixed before go-live
Medium	Feature works with a workaround or minor wrong behavior	Fixed before go-live or scheduled in warranty with client agreement
Low	Cosmetic, wording, layout	Fixed during warranty

11. Testing Strategy

Level	Tools	Coverage target
Unit	pytest, hypothesis (property tests for amortization and allocation)	90% on `services/` (financial logic), 80% overall backend
Integration	pytest + real MongoDB replica set in Docker (testcontainers)	All API endpoints, transactions, RBAC denials
Admin E2E	Playwright	Critical paths: login, register member, apply, approve, release, post payment, void, reports
Mobile E2E	Maestro (Android emulator and iOS simulator)	Register, login, view loan, apply, notifications
Performance	Locust	Section 8 targets with seeded data
Security	OWASP ZAP baseline scan, dependency audit (pip-audit, npm audit)	No High findings open at go-live
UAT	Client staff using the acceptance tests below	100% of Critical-priority tests pass

Property tests that must always hold: sum of installment principal = loan principal; final principal balance = ₱0.00; allocation never pays more than is due; balances never go negative; void + re-post returns the loan to the same state.

12. Acceptance Tests

Test data (sample settings, replaced in Phase 1):

- **Product A, "Regular Loan":** diminishing balance, 2% per month, monthly, terms 6–24 months, ₱5,000–₱200,000, processing fee 2% deducted from proceeds, penalty 3% per month of unpaid installment, 3 grace days, max 1 active loan.
- **Product B, "Quick Loan":** add-on, 1.5% per month, monthly, terms 3–6 months, ₱1,000–₱30,000, no fees, same penalty.
- **Member M1:** Juan Dela Cruz, active. **Member M2:** pending app registration.
- **Staff:** ADM (Admin), MGR (Manager), LO (Loan Officer), CSH (Cashier), AUD (Auditor).

Priority: **C** = Critical (must pass for go-live), **H** = High, **M** = Medium.

12.1 Authentication and Access (AT-AUTH, AT-RBAC)

ID	Pri	Test	Steps	Expected result
AT-AUTH-01	C	Staff login	Log in as LO with correct credentials	Dashboard opens; login recorded in audit trail with IP and time
AT-AUTH-02	C	Wrong password lockout	Enter wrong password 5 times for CSH	Account locked for 15 minutes; message shown; ADM can unlock; events audited
AT-AUTH-03	H	Session expiry	Leave admin idle past session timeout, then click any action	User is asked to log in again; no data is changed
AT-AUTH-04	C	Deactivated staff	ADM deactivates LO; LO tries to log in or uses an existing session	Login refused; existing session rejected on next request
AT-AUTH-05	H	Password policy	Set a new password "12345"	Rejected; minimum 8 characters with letters and numbers required
AT-RBAC-01	C	Cashier limits	Log in as CSH; try to open Loan Products, approve a loan, or call the approve API directly	Menus hidden; direct API call returns 403; attempt audited
AT-RBAC-02	C	Segregation of duties	LO encodes an application; LO (with an Approver role added for test) tries to approve the same application	Approval refused: "The encoder cannot approve this application"
AT-RBAC-03	H	Auditor read-only	Log in as AUD; open members, loans, payments, audit trail; try to edit any record	All viewable; no edit/create/void actions available; API writes return 403
AT-RBAC-04	C	Member data isolation	Log in to the app as M1; request M2's loan by changing the ID in the API call	404 returned; no data from another member is exposed

12.2 Members (AT-MEM)

ID	Pri	Test	Steps	Expected result
AT-MEM-01	C	Staff registers a member	LO fills required fields, uploads valid ID, saves	Member created as Active with a unique member number (e.g. M-2026-000001); audit entry created
AT-MEM-02	C	Member self-registration	On the app, M2 fills personal info, uploads ID, accepts privacy consent, verifies OTP	Account created as Pending; M2 sees "Awaiting verification"; staff sees it in Pending Registrations
AT-MEM-03	C	Approve app registration	LO reviews and approves M2	M2 becomes Active; M2 receives a push notification and can now apply for loans
AT-MEM-04	H	Duplicate check	Register another member with M1's mobile number, or same full name + birthdate	Mobile duplicate blocked; name+birthdate match shows a warning listing the possible duplicate

ID	Pri	Test	Steps	Expected result
AT-MEM-05	H	Required field validation	Save with missing birthdate, invalid mobile (e.g. 0917-12) or invalid email	Save blocked; each invalid field highlighted with a clear message
AT-MEM-06	H	Profile change approval	M1 changes address in the app	Request shown as pending to staff; profile updated only after staff approval; both versions in audit
AT-MEM-07	M	Search	Search by member number, partial last name, mobile	Correct members returned in under 2 seconds
AT-MEM-08	H	Suspended member	MGR suspends M1; M1 tries to apply for a loan	Application blocked with message; existing loans and payments still viewable
AT-MEM-09	H	Document privacy	Copy the URL of M1's ID image and open it after 10 minutes or in a logged-out browser	Access denied; links expire

12.3 Loan Products and Computation (AT-PRD)

ID	Pri	Test	Steps	Expected result
AT-PRD-01	C	Diminishing schedule	Preview Product A, ₱50,000, 12 months	Monthly installment **₱4,727.98**. Installment 1: interest ₱1,000.00, principal ₱3,727.98, balance ₱46,272.02. Installment 2: interest ₱925.44, principal ₱3,802.54. Installment 12: interest ₱92.71, principal ₱4,635.27, balance **₱0.00**. Total interest **₱6,735.76**, total payable **₱56,735.76**
AT-PRD-02	C	Add-on schedule	Preview Product B, ₱20,000, 6 months	Total interest **₱1,800.00**; installments 1–5 = ₱3,633.33 (principal ₱3,333.33 + interest ₱300.00); installment 6 = **₱3,633.35** (principal ₱3,333.35); total payable **₱21,800.00**
AT-PRD-03	C	Match client sheet	Enter 3 real past loans from the client's records	Every installment, total interest and balance matches the client's sheet to the centavo
AT-PRD-04	H	Product limits	Preview Product A for ₱250,000 or 30 months	Rejected with "Amount must be between ₱5,000 and ₱200,000" / "Term not allowed"
AT-PRD-05	C	Product change does not affect existing loans	Release a loan on Product A; change Product A rate to 2.5%	Existing loan schedule unchanged; new previews use 2.5%; change audited
AT-PRD-06	M	Deactivate product	MGR deactivates Product B	Not offered in app or admin for new applications; existing Product B loans unaffected
AT-PRD-07	H	Month-end due dates	Release a monthly loan on January 31	Due dates fall on Feb 28 (or 29), Mar 31, Apr 30, and so on

12.4 Loan Applications (AT-APP)

ID	Pri	Test	Steps	Expected result
AT-APP-01	C	Apply from mobile	M1 chooses Product A, ₱50,000, 12 months, sees schedule preview, uploads payslip, submits	Application number shown; status Submitted; staff dashboard shows it live without refresh; LO notified
AT-APP-02	C	Walk-in application	LO encodes an application for M1 in admin	Status Submitted, channel Walk-in; M1 sees it in the app

ID	Pri	Test	Steps	Expected result
AT-APP-03	C	Status tracking	LO marks Under Review; MGR approves with remarks	Each change appears in M1's app timeline with date and time; push notification sent for each
AT-APP-04	C	Decline	MGR declines with reason "Incomplete requirements"	M1 sees Declined with reason; notified; cannot be released
AT-APP-05	H	Member cancel	M1 cancels while Submitted; tries to cancel another that is Under Review	First succeeds; second blocked
AT-APP-06	C	Eligibility: max active loans	M1 has an active Product A loan; applies for another Product A loan	Blocked: "Maximum active loans for this product reached"
AT-APP-07	H	Required documents	Submit without a required document	Submission blocked; missing document named

12.5 Loan Release (AT-REL)

ID	Pri	Test	Steps	Expected result
AT-REL-01	C	Release with fees	MGR releases approved ₱50,000 Product A loan on 2026-12-01	Loan number generated; processing fee ₱1,000.00; net proceeds **₱49,000.00**; 12 installments due Jan 1 2027 to Dec 1 2027; status Active; release voucher printable; M1 notified
AT-REL-02	C	Atomic release	Simulate a database failure in the middle of a release	Nothing saved: no loan, no installments, application still Approved
AT-REL-03	C	Double release prevention	Click Release twice quickly, or two managers release the same application	Exactly one loan created; second attempt refused
AT-REL-04	C	Member sees loan	Open M1's app after release	Loan visible with balance ₱50,000.00 principal (₱56,735.76 total payable), next due Jan 1 2027 ₱4,727.98, 12 remaining installments

12.6 Payments (AT-PAY)

ID	Pri	Test	Steps	Expected result
AT-PAY-01	C	Exact installment payment	CSH posts ₱4,727.98 cash, OR 000101, for installment 1 of the AT-REL-01 loan, on time	Allocation: interest ₱1,000.00, principal ₱3,727.98; installment 1 Paid; principal balance ₱46,272.02; receipt printable; M1's app updates within 10 seconds
AT-PAY-02	C	Partial payment with penalty	Installment 1 unpaid; on day 10 after due (past 3-day grace), post ₱3,000.00	Penalty assessed ₱141.84. Allocation: penalty ₱141.84, interest ₱1,000.00, principal ₱1,858.16. Installment 1 Partial with ₱1,869.82 principal remaining
AT-PAY-03	C	Advance payment	Post ₱9,455.96 when only installment 1 is due	Installments 1 and 2 marked Paid; next due becomes installment 3
AT-PAY-04	C	Overpayment beyond balance	Post an amount greater than the total outstanding balance	Rejected with the exact payoff amount shown
AT-PAY-05	C	Full payoff	Post the payoff amount shown on the loan	All installments Paid; loan Fully Paid; balances ₱0.00; M1 notified; fully paid notice printable
AT-PAY-06	C	Duplicate OR number	Post a payment using OR 000101 again	Rejected: "OR number already used"
AT-PAY-07	C	Double-click / retry	Click Post twice or resend the same request (same idempotency key)	Exactly one payment recorded

ID	Pri	Test	Steps	Expected result
AT-PAY-08	C	Concurrent posting	Two cashiers post to the same loan at the same moment	Both succeed in sequence, or one is asked to retry; final balances are correct; no double allocation
AT-PAY-09	C	Void payment	MGR voids the AT-PAY-01 payment with reason "Wrong loan"	Payment status Voided (still visible, struck through); installment 1 back to Unpaid; balance restored to ₱50,000.00 principal; audit records who, when, why; M1 sees the correction
AT-PAY-10	C	Cashier cannot void	CSH tries to void a payment	Not allowed (403)
AT-PAY-11	H	Non-cash payment	Post via bank transfer with reference number	Method and reference saved and shown on receipt and reports
AT-PAY-12	H	Payment history	M1 opens Payment History	All posted payments listed newest first with date, OR number, amount and allocation; voided payments marked

12.7 Overdue and Monitoring (AT-OVD, AT-DSH)

ID	Pri	Test	Steps	Expected result
AT-OVD-01	C	Overdue status	Leave installment 1 unpaid; run the daily job for Jan 2, 2027	Installment Overdue; loan Past Due; days past due = 1; no penalty yet (within grace)
AT-OVD-02	C	Penalty after grace	Run the daily job for Jan 5 (day 4)	Penalty ₱141.84 assessed once
AT-OVD-03	C	Idempotent job	Run the daily job twice for the same date	Penalty not duplicated; numbers unchanged
AT-OVD-04	H	Second penalty period	Run for day 31 past due (still unpaid)	Second penalty period assessed per the rule (another ₱141.84)
AT-OVD-05	C	Aging buckets	Seed loans 0, 15, 45, 75, 120 days past due	Each appears in the correct bucket (Current, 1–30, 31–60, 61–90, 90+) with correct totals
AT-OVD-06	H	Penalty waiver	MGR waives ₱141.84 with reason	Penalty balance reduced; waiver in audit and Penalty report; LO cannot waive
AT-DSH-01	C	Dashboard totals	Compare dashboard figures with a manual sum of seeded data	Outstanding portfolio, releases, collections, due today, overdue amount all match exactly
AT-DSH-02	H	Live update	Keep dashboard open while CSH posts a payment on another computer	Collections today updates within 10 seconds without refreshing
AT-DSH-03	H	Due today / overdue list	Open Due Today and Overdue lists	Correct members, amounts, contact numbers; exportable

12.8 Notifications and Reminders (AT-NTF)

ID	Pri	Test	Steps	Expected result
AT-NTF-01	C	Before-due reminder	Installment due Jan 1; run reminder job Dec 29, 9:00 AM	M1 receives push: "Your payment of ₱4,727.98 is due on Jan 1, 2027"; stored in inbox
AT-NTF-02	C	Due-date and overdue reminders	Run on Jan 1 and Jan 2 with installment unpaid	Due-today and overdue reminders sent; no reminder if already paid
AT-NTF-03	H	No duplicates	Run the reminder job twice the same day	Only one reminder per installment per trigger
AT-NTF-04	H	Quiet hours	Trigger reminders at 10:00 PM	Held until 8:00 AM

ID	Pri	Test	Steps	Expected result
AT-NTF-05	C	Event notifications	Approve, decline, release, post payment, void payment	M1 receives the matching notification for each event
AT-NTF-06	M	Inbox read state	Open a notification	Marked read; unread badge decreases
AT-NTF-07	M	Push disabled	M1 disables push in phone settings	Notifications still appear in the in-app inbox

12.9 Mobile App Experience (AT-MOB)

ID	Pri	Test	Steps	Expected result
AT-MOB-01	C	Home summary	M1 opens app	Shows total outstanding, next due date and amount, overdue warning if any; values match admin
AT-MOB-02	C	Loan detail	Open the loan	Shows principal, interest and penalty balances, paid vs remaining installments (e.g. "3 of 12 paid, 9 remaining"), full schedule with status per installment
AT-MOB-03	H	Biometric unlock	Enable fingerprint/Face ID, close and reopen app	App unlocks with biometrics; falls back to password
AT-MOB-04	H	Offline viewing	Turn off internet and open app	Last synced data shown with "Last updated" time; actions that need internet are disabled with a message
AT-MOB-05	H	Devices	Run on a low-end Android 8 phone and an iPhone with iOS 15	All screens usable; no crashes; text readable
AT-MOB-06	C	Logout and token safety	Log out, then press back or reopen app	Login required; no member data visible

12.10 Real-Time Synchronization (AT-SYNC)

ID	Pri	Test	Steps	Expected result
AT-SYNC-01	C	Admin → app	With M1's app open, CSH posts a payment	App balance and history update within 10 seconds
AT-SYNC-02	C	App → admin	M1 submits an application	Appears in the admin Applications queue within 10 seconds with no refresh
AT-SYNC-03	H	Reconnect	Disconnect M1's phone, post a payment, reconnect	App shows updated data within 10 seconds of reconnecting

12.11 Reports (AT-RPT)

ID	Pri	Test	Steps	Expected result
AT-RPT-01	C	Collections report	Generate for a date range, by cashier	Totals equal the sum of posted (not voided) payments; per-cashier subtotals correct
AT-RPT-02	C	Outstanding balances	Generate as of a date	Each loan's balance matches its loan detail; grand total matches dashboard
AT-RPT-03	C	Aging report	Generate as of a date	Matches AT-OVD-05 results
AT-RPT-04	H	Releases report	Generate for a month	Lists every loan released with principal, fees, net proceeds
AT-RPT-05	C	Member ledger	Generate for M1	Chronological releases, payments, penalties, waivers and voids with running balance that ends at the current balance

ID	Pri	Test	Steps	Expected result
AT-RPT-06	H	Export formats	Export each report to PDF and Excel	Files open correctly; Excel numbers are numeric (summable), not text; PDF has company header, date range, generated by and time
AT-RPT-07	M	Large range	Generate a one-year collections report on seeded data	Completes in under 30 seconds

12.12 Security, Audit, Backup and Performance (AT-SEC, AT-AUD, AT-BKP, AT-PERF)

ID	Pri	Test	Steps	Expected result
AT-SEC-01	C	HTTPS only	Open admin over http://	Redirected to https://; valid certificate
AT-SEC-02	C	Passwords stored safely	Inspect the database	Passwords stored only as Argon2 hashes
AT-SEC-03	H	Injection and XSS	Enter `{"$ne": null}` in login fields and `<script>` in a name field	Login fails; script shown as plain text, never executed
AT-SEC-04	H	Vulnerability scan	Run OWASP ZAP baseline and dependency audit	No High or Critical findings open
AT-AUD-01	C	Audit completeness	Perform create member, edit member, approve, release, post, void, waive, settings change, login	Each action has an audit entry with user, time, IP, before and after values
AT-AUD-02	C	Audit tamper-proof	As ADM, try to edit or delete an audit entry via UI or API	Not possible
AT-BKP-01	C	Daily backup	Check backup storage after 2 days	Two encrypted backups present off-server with success logs; failure triggers an email alert
AT-BKP-02	C	Restore drill	Restore the latest backup to a fresh staging server	System runs; record counts and balances match the source as of backup time; restore completed within 4 hours
AT-PERF-01	H	Load	Locust: 200 concurrent users on seeded 50,000 members / 80,000 loans	p95 under 500 ms; no errors
AT-PERF-02	H	Daily job duration	Run the overdue job on seeded data	Completes in under 10 minutes

12.13 Usability (AT-UX)

ID	Pri	Test	Steps	Expected result
AT-UX-01	H	Cashier speed	A trained cashier posts a payment from member search to printed receipt	Done in under 1 minute, within 3 screens
AT-UX-02	H	First-time member	A member who has never used the app registers and finds their next due date without help	Completed without assistance
AT-UX-03	M	Confirmations	Try release, void, waive, suspend	Each asks for confirmation with a summary of the effect
AT-UX-04	M	Error messages	Trigger common errors	Messages are plain language and say what to do next

13. UAT Procedure and Sign-off

1. Staging is loaded with test data (Section 12) plus a copy of 20 to 50 real anonymized loans provided by the client.
2. Client staff (at least one Manager, one Loan Officer, one Cashier) run the tests over Week 9, recording Pass/Fail and remarks.
3. Defects are logged with screenshots and fixed daily; failed tests are re-run after fixes.
4. **Go-live criteria:** all Critical (C) tests pass, no open Critical or High defects, restore drill passed, client signs the UAT acceptance form.

Role	Name	Signature	Date
Client Representative			
Manager / Approver			
Developer	Jonah Pacas-Abanil, CpE		

14. Go-Live Checklist

- ❑ Production server hardened (firewall, SSH keys only, automatic security updates)
- ❑ MongoDB replica set with authentication, not publicly exposed; daily backup and alert verified
- ❑ Environment secrets set; debug off; CORS restricted to the admin domain
- ❑ Domain and SSL active; uptime and error monitoring active
- ❑ Loan products, penalty rules, settings and OR series configured with real values
- ❑ Staff accounts created with correct roles; default passwords changed
- ❑ Existing members and loans migrated and balances reconciled with client records
- ❑ Android app published on Google Play; iOS app submitted/approved on App Store under the company's accounts
- ❑ Staff trained; user guides delivered (Admin, Cashier, Member)
- ❑ Source code, credentials and runbook handed over
- ❑ 60-day warranty period start date recorded